Supplement to Rational Cycles of Periods Three and Four for Quadratic Hénon Maps

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Résumé. Ce supplément présente des démonstrations alternatives et des résultats auxiliaires pour les classifications de cycles rationnels de périodes 3 et 4 de l'article associé.
Abstract. This supplement gives alternative proofs and supporting results for the companion article's classifications of rational cycles of periods three and four.

References explicitly prefixed "Article" refer to the companion article; references S1–S6 are internal. The two infinitude arguments use finite fibers and ordered escape, respectively. The escape argument is independent of the six-parameter fiber bound. Selected formal proofs cover specific statements and do not formalize the entire article. Reproduction instructions and coverage accompany this supplement.

S1. Real multipliers and finite examples

The family is Article equation (6); all stability statements below refer only to that section. For use over $\mathbb{R}$, here is the direct verification of the section, independent of the arithmetic parametrization theorem.

Write $A = -5b - 7$, $B = 7b + 5$, $C = b - 1$ and $H = -29b^2 - 38b - 29$. Expansion gives the polynomial identities

$$B^2 + H + 4bA = 4C, \quad C^2 + H + 4bB = 4A, \quad A^2 + H + 4bC = 4B.$$

Dividing by 16 proves all three transitions. Moreover $x_1 - x_0 = 3(b+1) \neq 0$, so period 1 is excluded; primality of 3 proves minimality. At $b = -1$, the family itself has $c = -5/4$ and collapses to the fixed point $(-1/2, -1/2)$. Instead, at $c = -1$ the states $(0, -1), (-1, 0), (0, 0)$ form a 3-cycle by direct substitution and are distinct.

S1.1. Multipliers and the exceptional parameters. The polynomial identities and nonfixity argument above also hold over $\mathbb{R}$, as does the inverse formula for $b \neq 0$. We use "hyperbolic saddle" to mean one multiplier has

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modulus strictly less than 1 and the other strictly greater than 1. At the starting state (x_0, x_1) the chain-rule order is

$$M_b = J_b(x_0)J_b(x_2)J_b(x_1), \qquad J_b(v) = \begin{pmatrix} 0 & 1 \\ b & 2v \end{pmatrix}.$$

Proposition S1.1 (Real saddle classification)**.** *For every $b \in \mathbb{R} \setminus \{-1, 0, 1\}$, the cycle in Article equation* (6) *is a hyperbolic saddle. At $b = 1$ it is not hyperbolic. At $b = 0$ it is a cycle of an endomorphism with multipliers $0, 35/8$. At $b = -1$ the family has minimum period 1.*

Proof. Multiplication yields

$$M_b = \begin{pmatrix} b(b-1)/2 & (b+1)(7b-5)/4 \\ -b(b+1)(5b-7)/4 & -(b-1)(35b^2+66b+35)/8 \end{pmatrix}.$$

Thus $T_b = \operatorname{tr} M_b = -(b-1)(35b^2+62b+35)/8$, $\det M_b = -b^3$, and $p_b(\lambda) = \lambda^2 - T_b\lambda - b^3$ satisfies

$$p_b(1) = \frac{27(b-1)(b+1)^2}{8}, \qquad p_b(-1) = -\frac{(b-1)(43b^2+70b+43)}{8}.$$

The last quadratic is $43(b + 35/43)^2 + 624/43 > 0$. For $b \neq \pm 1$, the two endpoint values have opposite signs. The intermediate value theorem gives a root in $(-1, 1)$; since the polynomial is monic and negative at one endpoint, it has another root outside $[-1, 1]$. This proves the strict inequalities, not merely reality of the roots. The determinant is nonzero if $b \neq 0$.

At $b = 1$, direct substitution gives $M_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, with multipliers $-1, 1$. At $b = 0$, $p_0(\lambda) = \lambda(\lambda - 35/8)$. At $b = -1$, the derivative of the minimum return (one iterate) is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & -1 \end{smallmatrix}\right)$, with roots $(-1 \pm i\sqrt{3})/2$ on the unit circle; $M_{-1} = I$ is the derivative of three iterates of that fixed point. No nonlinear stability follows from these neutral multipliers. This classification does not address the separate cycle at $b = -1, c = -1$. □

S1.2. Examples of rational periods. The following words give rational cycles. Each entry can be verified by substitution in Article equation (2); excluding the proper divisors of its stated length proves minimum period. For period 4 the word has $x_2 \neq x_0$; for period 6 it has $x_1, x_2, x_3 \neq x_0$; for period 8 it has $x_1, x_2, x_4 \neq x_0$. For the prime lengths 2, 3 and 7, nonfixity suffices. These are existence statements, without an exclusion of other periods. Ingram already records attainment of all six listed periods at $b = 1$ [2, p. 4]. No priority is asserted for the representatives in this table. The period-8 orbit is Ingram's orbit, with a different starting point, and contains his point $(1/4, -3/4)$ at $c = -9/16$ [2, p. 4].

b	n	c	Scalar word
1	1	$-1/4$	$(-1/2)$
1	2	-4	$(-2, 2)$
1	3	-6	$(-3, 3, 0)$
1	4	$-5/4$	$(-3/2, 3/2, -1/2, 1/2)$
1	6	$-7/9$	$(-1/3, 2/3, -2/3, 1/3, -4/3, 4/3)$
1	8	$-9/16$	$(-5/4, 5/4, -1/4, 3/4, -1/4, 1/4, -3/4, 1/4)$
-1	7	$-9/16$	$(-3/4, -1/4, 1/4, -1/4, -3/4, 1/4, 1/4)$

For instance, multiplying the period-7 coordinates by 4 gives

$$A = (-3, -1, 1, -1, -3, 1, 1).$$

The seven integer identities are $4A_{i+2} = A_{i+1}^2 - 9 - 4A_i$. Their right sides are $(4, -4, -12, 4, 4, -12, -4)$, proving closure. Nonfixity and primality of 7 prove minimum period. All seven state pairs are distinct and the odd numerators give reduced denominator 4. The period-7 and period-8 examples at different b do not alone prove that the full rational period sets differ.

S2. The period-three curve at $b = 1$

We recall a plane curve that records the first scalar x and the parameter c of a marked rational 3-cycle. Its relation to the dynamics, including the exceptional points of the projection, is essential to the parametrization.

Write the word as (x, y, z) and put $s = x^2 + c$. The cycle equations at $b = 1$ are

$$z = y^2 + c + x, \qquad x = z^2 + c + y, \qquad y = x^2 + c + z.$$

The last equation gives $y - z = s$. Substituting $y = z + s$ and $c = s - x^2$ in the other two yields

$$z^2 + z + 2s - x^2 - x = 0, \qquad z^2 + (2s - 1)z + s^2 + s - x^2 + x = 0.$$

Their difference is $2(s - 1)z + s^2 - s + 2x = 0$. If $s = 0$, then $y = z$ and subtraction of the cycle equations forces $x = y$, so the word is fixed. If $s = 1$, the difference forces $x = 0$ and the first quadratic is $z^2 + z + 2 = 0$, with discriminant -7; no rational cycle occurs. Thus every rational primitive word has $s \neq 0, 1$, and

$$z = -s/2 - x/(s - 1), \qquad y = s/2 - x/(s - 1).$$

Substitution in the first quadratic gives $sG(x, s)/(4(s - 1)^2)$, where

$$G(x, s) = (s - 1)^2(s + 6) - 4x^2(s - 2). \tag{1}$$

Consequently $G = 0$ is necessary, and on $s \neq 0, 1$ the reconstruction proves sufficiency for the three return equations. The invertible affine change $s = x^2 + c$ gives the plane equation

$$F(x, c) = x^6 + 3x^4c + 3x^2c^2 + 4x^2c - 3x^2 + c^3 + 4c^2 - 11c + 6 = 0. \tag{2}$$

This sextic is exactly the specialization of Ross's primitive-period-3 polynomial [3, §3.1, p. 54]. His convention is $H_{a,B}(u,v) = (a - u^2 - Bv, u)$: the change $T(x,y) = (-y,-x)$ gives $H_{a,B} \circ T = T \circ f_{c,b}$ for $a = -c$, $B = -b$. Setting $B = -1$ and $u = -x$ in that polynomial gives (2). The equation is cited from Ross; the reconstruction below is retained because it specifies the marked lifting and its exceptions. The following parametrization also proves primitivity and identifies the rational image completely.

Proposition S2.1 (Rational parametrization and infinitude)**.** *The curve* (2) *is geometrically integral and its projective normalization is* $\mathbb{P}^1_{\mathbb{Q}}$*, of geometric genus zero. There are infinitely many* $c \in \mathbb{Q}$ *for which* $f_{c,1}$ *has a rational point of minimum period 3.*

Proof. Over $\overline{\mathbb{Q}}(s)$ the function $(s-1)^2(s+6)/(4(s-2))$ is not a square, since its order at $s = -6$ is 1. The quadratic equation for x is irreducible over that field; its coefficients are relatively prime in $\overline{\mathbb{Q}}[s]$. Gauss's lemma proves geometric integrality. On $s \neq 1$ put $t = x/(s-1)$ and $D = 4t^2 - 1$. Equation (1) gives

$$s = \frac{8t^2+6}{D}, \qquad x = \frac{t(4t^2+7)}{D}, \qquad c = -\frac{16t^6+24t^4+33t^2+6}{D^2}. \tag{3}$$

These formulas and $t = x/(s-1)$ are inverse on dense opens (exclude $D(4t^2+7) = 0$ over $\overline{\mathbb{Q}}$). The function field is $\mathbb{Q}(t)$; the smooth projective normalization is therefore $\mathbb{P}^1_{\mathbb{Q}}$.

For rational $t \neq \pm 1/2$, let $y = s/2 - t$ and $z = -s/2 - t$. Substitution in (3) gives

$$z = y^2 + c + x, \qquad x = z^2 + c + y, \qquad y = x^2 + c + z.$$

The polynomial identities can be checked by clearing D^2. Further,

$$y - x = -\frac{4t^2+3}{2t+1} \neq 0 \quad (t \in \mathbb{Q} \setminus \{\pm 1/2\}).$$

Thus these states have minimum period 3. For any fixed c_0, the equation $c(t) = c_0$ becomes a nonzero degree-6 polynomial, with leading coefficient -16, after multiplication by D^2. Each fiber has at most six rational parameters. There are infinitely many admissible rational t, so infinitely many distinct $c(t)$. This is a finite-fiber argument, not a numerical search. □

The parametrization concerns $b = 1$ and minimum period 3. Section S2.1 identifies the points omitted by the affine chart and extends the map to the projective normalization. Section S4 uses this family in a second proof of infinitude for every fixed rational b.

S2.1. The normalization and its exceptional points. The affine point $(x, c) = (0, 1)$ is exceptional. Indeed $s = 1$ in (1) forces $x = 0$; $s = 2$ gives $G = 8$, so is impossible. For rational t, $4t^2 + 7 \neq 0$. Hence (3) maps $\mathbb{Q} \setminus \{\pm 1/2\}$ bijectively onto the rational points of (2) other than $(0, 1)$. The correspondence with dynamics keeps the initial state (x, y) marked; it is not a bijection with unmarked orbits or with values of c.

At $s = 1$, the two quadratics in Section S2 force $x = 0$ and $z^2 + z + 2 = 0$. Its discriminant is -7, so the rational singular point has no rational lift. At $s \neq 1$, fixed degenerations satisfy $y = x = z$, whence $s = 0$ and $t^2 = -3/4$; none is rational. Every rational minimum-period-3 point therefore falls in the open already described.

The projective extension of the parametrization, useful for checking poles, is

$$[T : U] \longmapsto [TU(4T^2 + 7U^2)(4T^2 - U^2) : N_h : U^2(4T^2 - U^2)^2],$$

where $N_h = -16T^6 - 24T^4U^2 - 33T^2U^4 - 6U^6$. These forms have no common zero: outside $U(4T^2 - U^2) = 0$ the last coordinate is nonzero; at $U = 0$ the middle coordinate is $-16T^6$; at $4T^2 = U^2$ it is $-16U^6$. This nonconstant morphism from the projective line to the integral projective curve is finite. Since it is birational with normal source, it is the normalization. The parameters $\infty, \pm 1/2$ map to $[0 : 1 : 0]$ and are not affine cycles. The points above $(0, 1)$ satisfy $4t^2 + 7 = 0$ and are nonrational.

S3. Direct difference derivation for period 3

This is the alternative derivation of Article Proposition 2.1, including its fiber proof. The symbols in Article equations (3) and Article equation (4) are used below.

The three closure equations are

$$z = y^2 + c + bx, \qquad x = z^2 + c + by, \qquad y = x^2 + c + bz.$$

If $x = y$, subtracting the first and third gives $(b+1)(z-x) = 0$. For $b \neq -1$, any equality between two scalars therefore forces fixity, by rotation. A primitive word has $d = y - x \neq 0$ and $r = (z - x)/d \notin \{0, 1\}$. Equating the three expressions for c and dividing by d gives

$$(r - 1)(2x + (r + 1)d) = -r - b, \qquad r(2x + rd - b) = -(b + 1).$$

The determinant for the unknowns $(2x, d)$ is $-r(r-1)$. Solving gives Article equation (3). Conversely these equations imply all three expressions for c agree, so the formulas close the cycle. Since $4q_r = (2r - 1)^2 + 3 > 0$ and $b + 1 \neq 0$, we have $d \neq 0$ and hence minimum period 3. The ratio recovers r uniquely. No division by b is used.

For infinitude with this b fixed, write

$$\begin{aligned} L &= 2r(r-1), \\ X &= br^3 - br + b + r^3 - r^2 + 1, \\ Y &= br^3 - 2br^2 + br - b + r^3 - 3r^2 + 2r - 1, \\ Z &= -(br^3 - 2br^2 + 3br - b + r^3 - r^2 + 2r - 1). \end{aligned}$$

Then $(x, y, z) = (X/L, Y/L, Z/L)$ and $c = N/L^2$, where $N = ZL - Y^2 - bXL$. The polynomial Y has leading coefficient $b+1$ in degree 3; the other terms in N have degree at most 5. Consequently $N - c_0L^2$ has degree 6 with leading coefficient $-(b+1)^2 \neq 0$, for every $c_0 \in \mathbb{Q}$. Each fiber has at most six rational parameters. For finite A, at most $6|A|$ parameters map into A; one of the $6|A|+1$ integers $2, \ldots, 6|A|+2$ thus yields $c \notin A$.

At $b = -1$, three distinct scalars would permit the same difference calculation and force $d = 0$, a contradiction. After rotation put $x = y = a$, $z = e$. Closure reduces to $c = e - a^2 + a$ and $(e-a)(e+a+1) = 0$. The first factor gives fixity; the second gives Article equation (4). Its three states are distinct exactly when $a \neq -1/2$. Removing that one a does not remove its value $c = -5/4$, since $a = 1/2$ gives the primitive word $(1/2, 1/2, -3/2)$ at the same c. Finally $a = n$, for integers $n \geq 0$, gives $c_n = -1 - n^2$ and $c_n - c_m = (m-n)(m+n) > 0$ whenever $m > n$. These distinct values prove the assertion for every finite A also in this stratum.

S4. A second infinitude argument by transport and escape

A different proof uses the marked lifting at $b = 1$ from Section S2 and Section S2.1, instead of the degree-six fiber bound. Let (v_0, v_1, v_2) be its word, let $q = c(t)$ in (3), and put $\sigma = v_0 + v_1 + v_2$. For $b \neq -1$ set

$$\alpha = \frac{b+1}{2}, \quad \beta = \frac{b-1}{4}, \quad u_i = \alpha v_i + \beta, \quad C = \alpha^2 q - 2\alpha\beta\sigma - 5\beta^2. \tag{4}$$

For each cyclic index i, expansion gives the identity

$$u_{i+2} - u_{i+1}^2 - C - bu_i = \alpha^2(v_{i+2} - v_{i+1}^2 - q - v_i).$$

Indeed all three linear coefficients in the difference are $-2\alpha\beta$ and cancel the $2\alpha\beta\sigma$ term; the constant cancels because $b - 1 = 4\beta$. Since $\alpha \neq 0$, the inverse $v_i = (u_i - \beta)/\alpha$ and $q = (C + 2\alpha\beta\sigma + 5\beta^2)/\alpha^2$ is rational. Thus closure and nonfixity are preserved in both directions. This transports words together with their parameters. It is not a conjugation of the planar maps: C depends on the word through σ. It supplies no stability conclusion for these new families. The two rational charts are related by $r = (2t+1)/(2t-1)$, with inverse $t = (r+1)/(2(r-1))$.

For real $t \geq 1$, $D = 4t^2 - 1 \geq 3$ and direct simplification of (3) gives

$$q = -t^2 - \frac{32t^4 + 32t^2 + 6}{D^2} < -t^2, \qquad \sigma = -t + \frac{8t}{D}, \qquad |\sigma| < 4t.$$

Therefore $C < -\alpha^2 t^2 + 8|\alpha\beta|t - 5\beta^2$. If $w = |\alpha|t \geq 16|\beta| = 16v$, then

$$2(-w^2 + 8vw - 5v^2) + w^2 = -w(w - 16v) - 10v^2 \leq 0,$$

so $C < -\alpha^2 t^2/2$. For any real bound M, choose an integer t large enough to meet this threshold and $\alpha^2 t^2/2 > -M$. This gives a rational primitive cycle with $C < M$. Taking $M = \min A$ for nonempty finite A proves escape from A; the empty case is immediate. The argument includes $b = 0$ ($\alpha = 1/2$, $\beta = -1/4$). At $b = -1$ the separate sequence in the proof above applies, or one can use the at-most-two fibers of $a \mapsto -1 - a^2$.

S5. Denominator criterion and limits of finite computation

Fix an integer $q \geq 2$, $b = 1$, $c = -(2q+1)/q^2$, and seeds $x_0 = -(q+1)/q$, $x_1 = (q+1)/q$. Define $x_{n+2} = x_{n+1}^2 + c + x_n$ for $n \geq 0$, and $a_n = qx_n$. The scaled recurrence is

$$a_{n+1} = (a_n^2 - 1)/q + a_{n-1} - 2 \qquad (n \geq 1).$$

All x_n, $n \geq 0$, have positive reduced denominator exactly q if and only if all a_n are integers and $a_n^2 \equiv 1 \pmod{q}$ for every $n \geq 0$. In the forward direction, the recurrence gives the congruence for $n \geq 1$ and the seed gives it for $n = 0$. Conversely the congruences imply $\gcd(a_n, q) = 1$, so no denominator cancels. The equivalence does not assert that either side holds for an arbitrary q; it does not prove periodicity, closure or minimum period.

For $q = 2, 3, 4$, the words of Section S1.2 close with periods $4, 6, 8$ (up to a cyclic change of the seed for period 6). At $q = 5$, the sequence starts $a_0 = -6, a_1 = 6, a_2 = -1, a_3 = 4, a_4 = 0, a_5 = 9/5$, so $x_4 = 0$ already fails reduced denominator 5 and $x_5 = 9/25$. This defeats that denominator pattern at $q = 5$. It is not a proof of absence of period 10 for arbitrary seeds, nor a proof that the construction fails for every $q \geq 5$. The density of residues satisfying a congruence is not a probability law for this deterministic orbit.

S6. Alternative quotient verification and auxiliary period-four statements

These results concern the normalizations of the components in Article Section 3. They do not classify the full singular quotient scheme of their union.

S6.1. Rotation quotients and rational lifting. Let ρ rotate a word by one place. On L_ε the rotation sends (ε, t) to $(-\varepsilon, \varepsilon t)$, so ρ^2 sends t to $-t$. An invariant rational function on $L_+ \cup L_-$ is determined by its restriction to L_+, which must be even. With $a = t^2$, writing $k(t) = k(a) \oplus tk(a)$ and comparing the two coefficients under $t \mapsto -t$ shows that the invariant algebra in $k(t) \times k(t)$ is the field $k(a)$. Conversely each function of a extends invariantly across the two components. The normalized quotient is $\mathbb{P}^1_k$; an affine $a \in \mathbb{Q}$ lifts to a rational primitive marking exactly when $a \in (\mathbb{Q}^*)^2$, and then $c = -1 - a$. A rational quotient point alone is insufficient, for example $a = -1$.

On $\mathcal{C}$, rotation is

$$\rho(r) = \frac{r+1}{1-r}, \qquad \rho^2(r) = -1/r, \qquad v = S - 4/S = \frac{r^4 - 6r^2 + 1}{r(r^2-1)}.$$

This is the classical cyclic quartic action and invariant field in Hoshi [1, §3, p. 7]: his generator $(r-1)/(r+1)$ is ρ^{-1} and generates the same group. In particular, its invariant field $k(v)$ and the quartic below are known. Here $v = x + y + z + w$ is the sum of the Hénon word and $c = -3 - v^2/4$. For completeness, we give a direct verification using two quadratic fixed spaces. Put $\sigma = \rho^2$. Since $r^2 - Sr - 1 = 0$ and $S^2 + 4$ is not a square in $k(S)$ (it has simple zeros over $\bar{k}$), $k(r) = k(S) \oplus rk(S)$. The equality $\sigma(r) = S - r$ forces the coefficient of r in a fixed element to vanish, because $(2r - S)b(S) = 0$ implies $b = 0$. Hence $k(r)^\sigma = k(S)$. The residual involution sends S to $-4/S = v - S$. Now $S^2 - vS - 4 = 0$ and $v^2 + 16$ is not a square in $k(v)$ for the same simple-zero reason. Comparing coefficients in $k(S) = k(v) \oplus Sk(v)$ proves $k(r)^{\langle\rho\rangle} = k(v)$. Thus the normalized quotient is again $\mathbb{P}^1_k$.

For a given $v \in \mathbb{Q}$, a rational marking exists if and only if there are $S \in \mathbb{Q}^*$ and $U \in \mathbb{Q}$ with

$$S^2 - vS - 4 = 0, \qquad U^2 = S^2 + 4.$$

Necessity follows from $S = x + z$, $U = w - y$; sufficiency follows by substitution in Article equation (9), since $U \neq 0$ over $\mathbb{Q}$. Equivalently there must be a rational root of

$$r^4 - vr^3 - 6r^2 + vr + 1 = 0.$$

Indeed $r = (S + U)/2$ recovers $S = r - 1/r$ and conversely; such a rational root cannot be $0, \pm 1$. The value $v = 0$ does not lift rationally: it forces $S^2 = 4$, $U^2 = 8$. The boundary maps to $v = \infty$; the nonprimitive geometric parameters $r = \pm i$ map to $v^2 + 16 = 0$, as follows from $v^2 + 16 = (S^2 + 4)^2/S^2$. This describes quotients of normalizations with explicit arithmetic lifting conditions, not all rational points of a singular orbit space.

S6.2. Infinitude from the conic component alone. The function $c_{\mathcal{C}}(r)$ in Article equation (9) has at most eight parameters in any fixed c fiber: its cleared fiber polynomial has degree 8 with leading coefficient 1, independently of c. Since $\mathbb{Q} \setminus \{0, \pm 1\}$ is infinite, this component alone supplies infinitely many distinct c.

An argument avoiding fiber counts is also available. For real $r \geq 2$, put $D_4 = 4r^2(r^2-1)^2$ and $E = 2r^6 + 13r^4 + 1$. Then

$$c_{\mathcal{C}}(r) = -r^2/4 - E/D_4, \qquad 3D_4 - E = r^4(10r^2 - 37) + 12r^2 - 1 > 0.$$

Since $D_4, E > 0$, this gives $-r^2/4 - 3 < c_{\mathcal{C}}(r) < -r^2/4$. Choose $r_n = 4n$, $n \geq 1$. For $m > n$, $(r_m^2 - r_n^2)/4 = 4(m-n)(m+n) \geq 12 > 3$, so

$$c_{\mathcal{C}}(r_m) < -r_m^2/4 < -r_n^2/4 - 3 < c_{\mathcal{C}}(r_n).$$

The rational primitive lifts are already supplied by Article equation (9); the sequence proves infinitude with those hypotheses, without a genus or fiber-degree argument.

S6.3. Uniform count for each fixed map. For every $c_0 \in \bar{k}$ there are at most 12 distinct geometric points of minimum period 4, and therefore at most three orbits. One proof counts at most $2 + 2 + 8$ parameters in the three normalized families; intersections can only reduce the number of points. The degrees remain bounds after every specialization of c.

A second proof handles possible nonreduced fibers directly. The full return algebra is

$$B = \bar{k}[x, y, z, w]/(x^2 - y + w + c_0,\ y^2 - z + x + c_0, z^2 - w + y + c_0,\ w^2 - x + z + c_0).$$

Write its generators $g_i = X_i^2 - L_i$, where each L_i is linear or constant. For every pair,

$$X_j^2 g_i - X_i^2 g_j = L_j g_i - L_i g_j.$$

In a graded monomial order the terms on the right have degree at most 3, below the least common multiple $X_i^2 X_j^2$ of degree 4. The leading monomials are pairwise coprime, so the Buchberger criterion [4, Theorem 2.1.8] gives a Gröbner basis. All generators are monic for every c_0. The sixteen square-free monomials are therefore a basis of B.

The lower-period quotient, obtained by setting $z = x$ and $w = y$, is $\bar{k}[x, y]/(x^2 + c_0, y^2 + c_0)$, of dimension 4, even at $c_0 = 0$. Decompose the finite algebra B into its local Artinian factors. The quotient is supported on the lower-period factors, whose total length is consequently at least 4. The remaining factors, where $z - x$ is invertible, have total length at most $16 - 4 = 12$. Each distinct primitive point contributes at least one to that length. This proves the bound also in degenerate fibers; subtracting only the number of lower-period points would not suffice.

At $c = -3$ there are exactly four geometric primitive points, at the intersections $L_\varepsilon \cap \mathcal{C}$: $S^2 = 4$, $U^2 = 8$. At $c = 0$, the two lines give four

distinct points, and the conic fiber polynomial is $p(r) = r^8 + 14r^4 + 1$. It has eight simple roots: $p'(r) = 8r^3(r^4 + 7)$ has no common root with p, since $p(0) = 1$ and substituting $r^4 = -7$ gives -48. None is $0, \pm 1, \pm i$, and the components intersect only at $c = -3$. Thus the bound 12 is attained geometrically. Neither of these fibers has a rational primitive point, as proved in Article Proposition 3.2. A bound for each fixed map does not imply finiteness of parameters c.

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